# Dodgers Attendance Analysis – 2022 Season

This analysis explores the factors influencing attendance at Los Angeles Dodgers home games in the 2022 season. The goal is to provide actionable recommendations to increase fan turnout.

In [33]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load the data
dodgers_df = pd.read_csv('/Users/bryanemer/Downloads/dodgers-2022.csv')
dodgers_df.head()
```

Out[33]:

| | month | day | attend | day_of_week | opponent | temp | skies | day_night | cap | shirt | fireworks | bobblehead |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | APR | 10 | 56000 | Tuesday | Pirates | 67 | Clear | Day | NO | NO | NO | NO |
| 1 | APR | 11 | 29729 | Wednesday | Pirates | 58 | Cloudy | Night | NO | NO | NO | NO |
| 2 | APR | 12 | 28328 | Thursday | Pirates | 57 | Cloudy | Night | NO | NO | NO | NO |
| 3 | APR | 13 | 31601 | Friday | Padres | 54 | Cloudy | Night | NO | NO | YES | NO |
| 4 | APR | 14 | 46549 | Saturday | Padres | 57 | Cloudy | Night | NO | NO | NO | NO |

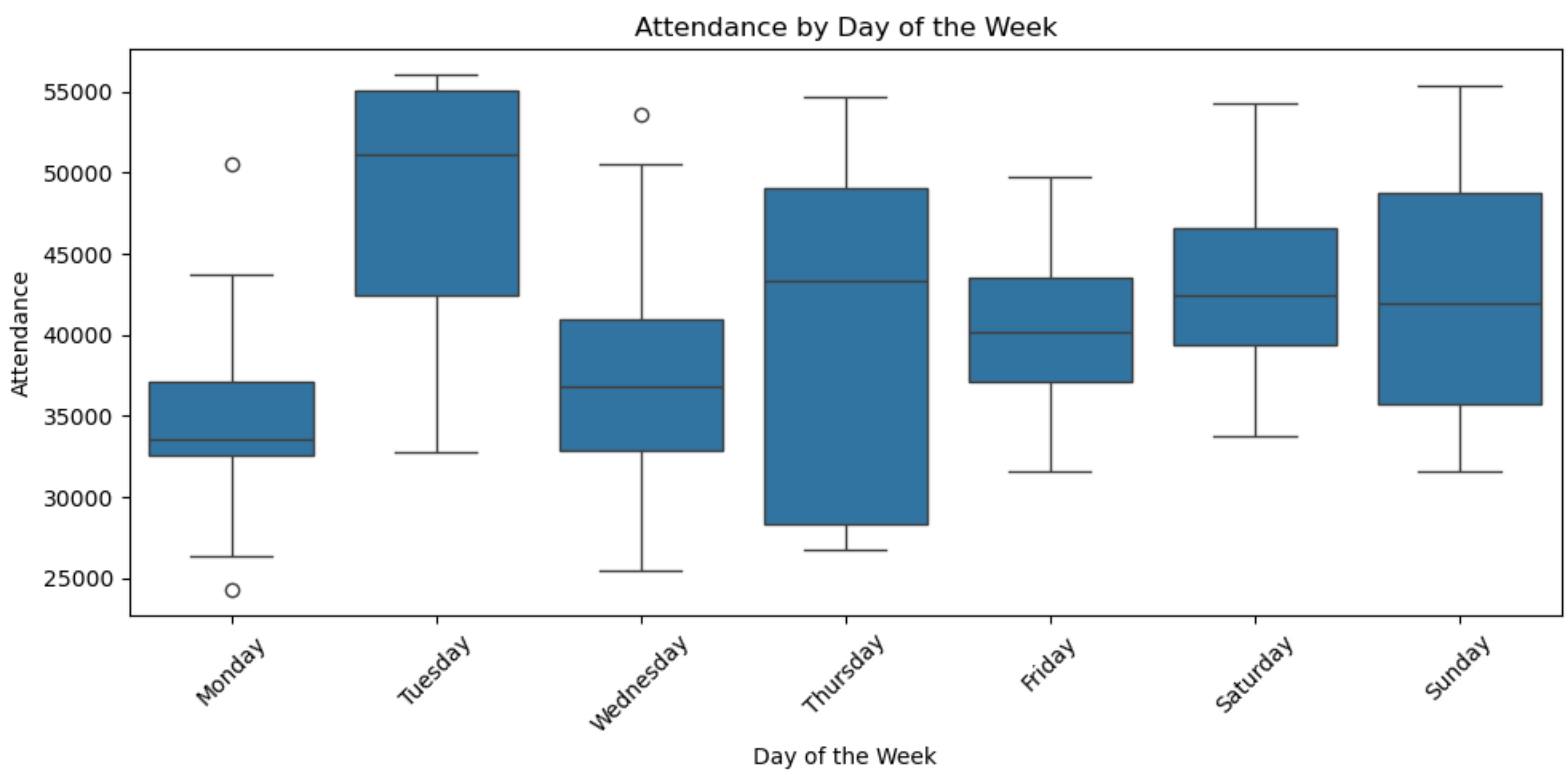
The dataset includes 81 home games for the Dodgers' 2022 season, including the following features:

- Date and weather: month, day, temp, skies, day_night
- Opponent and schedule: day_of_week, opponent
- Promotions: cap, shirt, fireworks, bobblehead
- Attendance: attend

In order to provide a recommendation for improving attendance, I will explore attendance patterns based on different factors, visualize key trends, and summarize insights to make my final recommendation.

## Attendance by Day of the Week

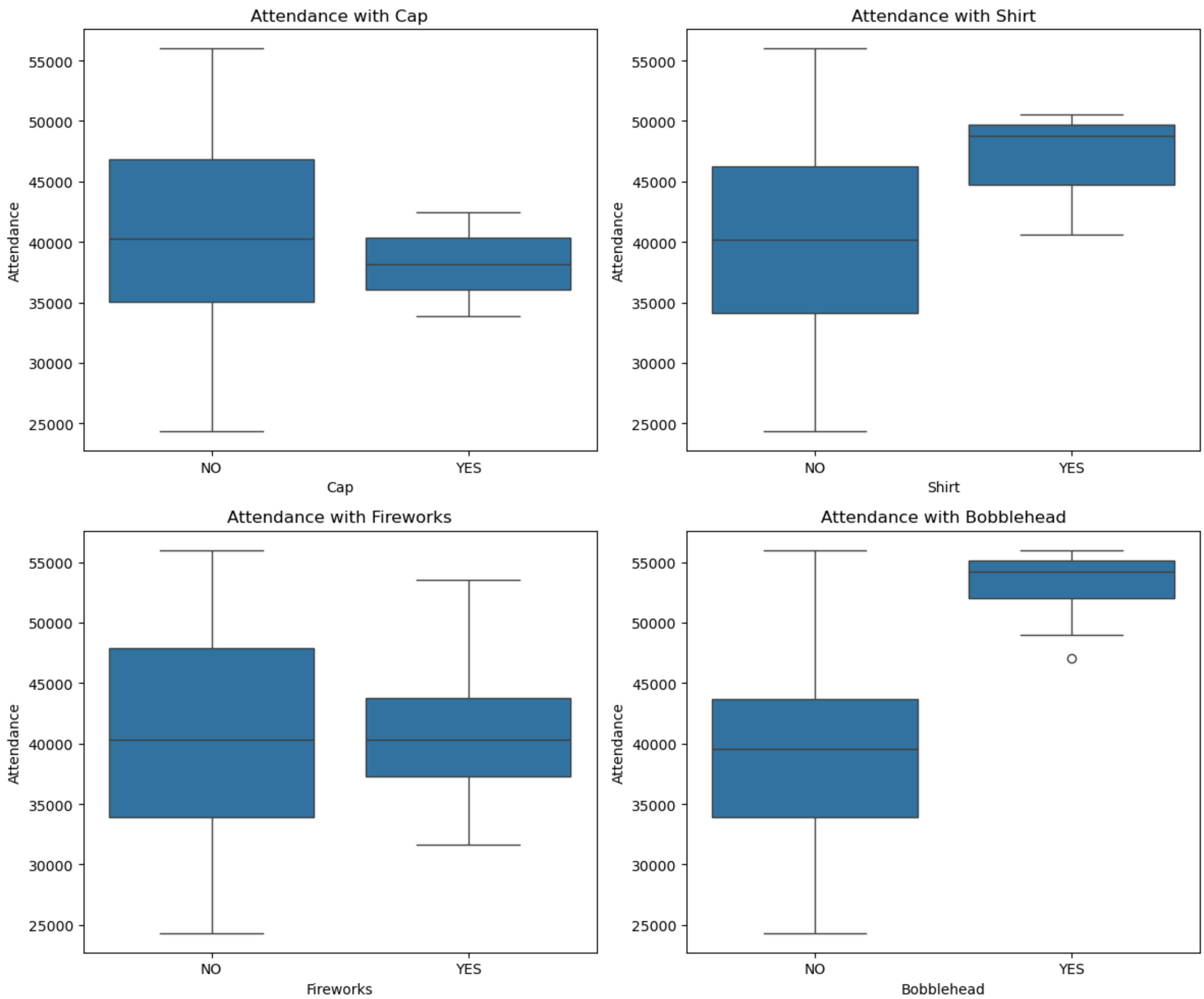
In [34]:
```
plt.figure(figsize=(10, 5))
sns.boxplot(x="day_of_week", y="attend", data=dodgers_df,
            order=["Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"])
plt.title("Attendance by Day of the Week")
plt.ylabel("Attendance")
plt.xlabel("Day of the Week")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```




We can see from the data that weekend games have higher attendance on average than weekday games, with the exception of Tuesday. Monday appears to have particularly low average attendance.

## Attendance and Promotional Events

In [35]:
```
promo_features = ['cap', 'shirt', 'fireworks', 'bobblehead']
fig, axes = plt.subplots(2, 2, figsize=(12, 10))
axes = axes.flatten()
for i, feature in enumerate(promo_features):
    sns.boxplot(x=feature, y="attend", data=dodgers_df, ax=axes[i])
    axes[i].set_title(f"Attendance with {feature.capitalize()}")
    axes[i].set_ylabel("Attendance")
    axes[i].set_xlabel(f"{feature.capitalize()}")
plt.tight_layout()
plt.show()
```



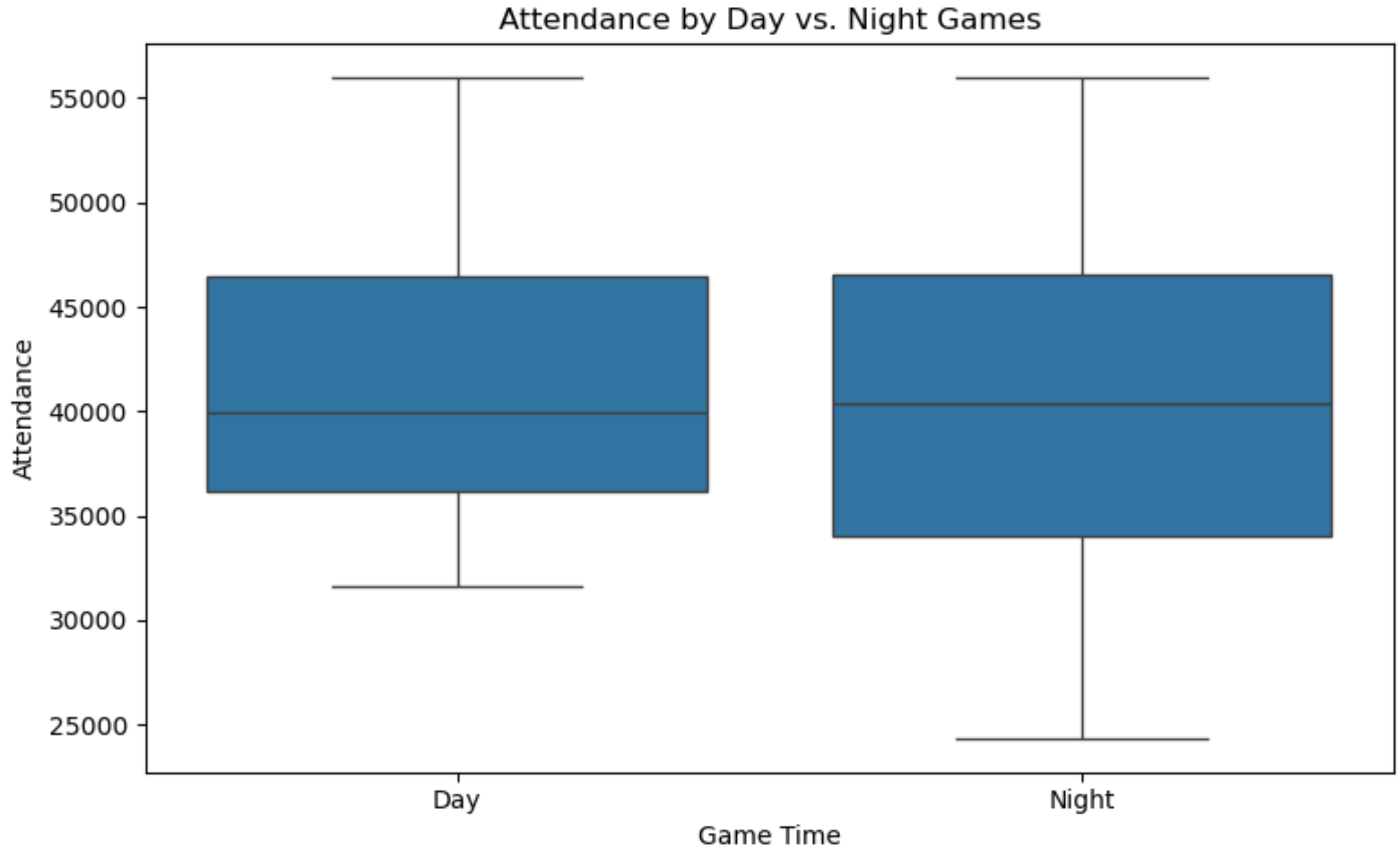

From the data we can see that while cap giveaways and fireworks do not seem to have much impact on attendance, we see higher average attendance with shirt giveaways and particularly bobblehead giveaways.

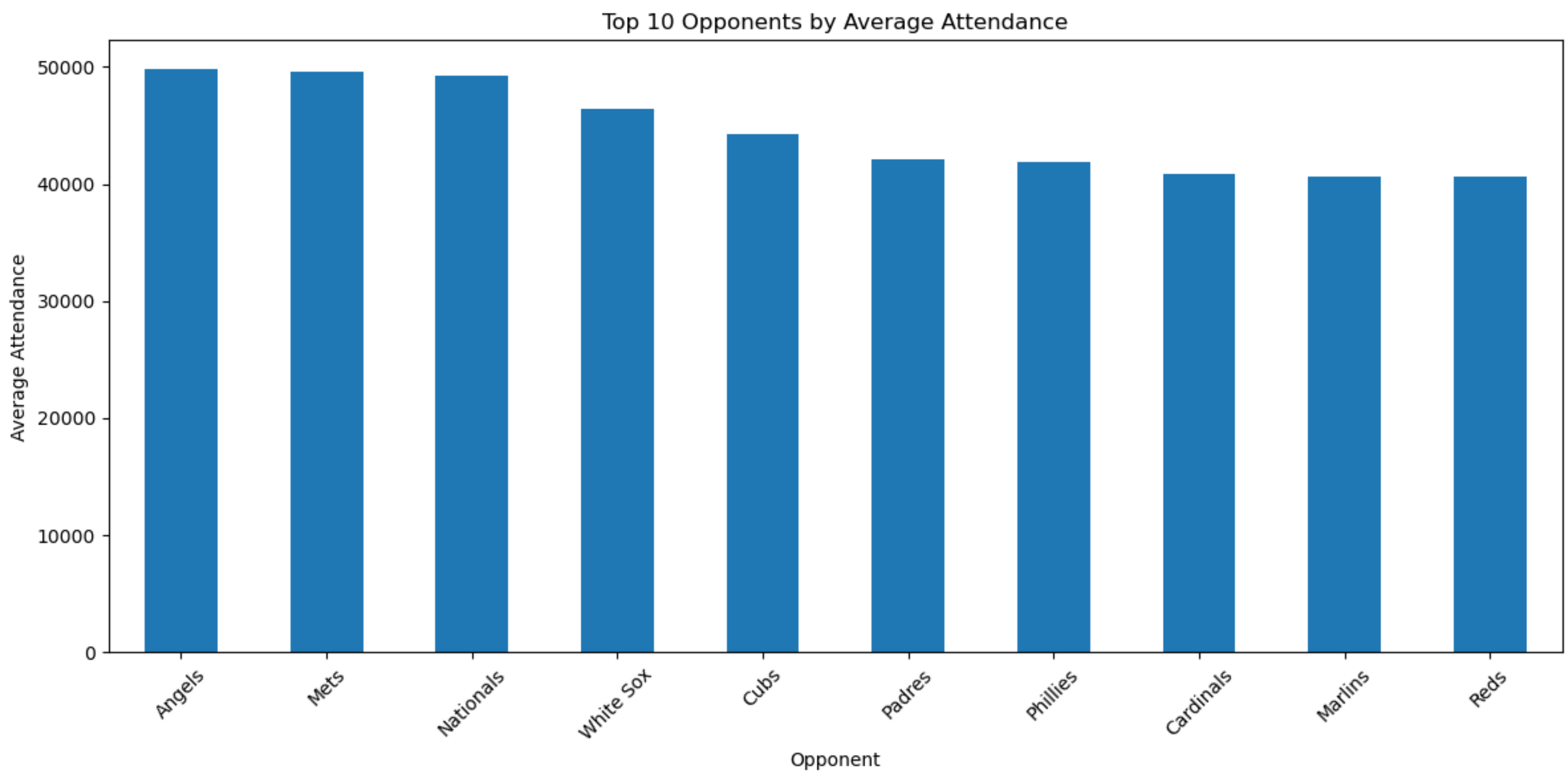
## Top Opponents and Game Timing

In [36]:
```
top_opponents = dodgers_df.groupby("opponent")["attend"].mean().sort_values(ascending=False)

plt.figure(figsize=(8, 5))
sns.boxplot(x="day_night", y="attend", data=dodgers_df)
plt.title("Attendance by Day vs. Night Games")
plt.ylabel("Attendance")
plt.xlabel("Game Time")
plt.tight_layout()
plt.show()

plt.figure(figsize=(12, 6))
top_opponents.head(10).plot(kind='bar')
plt.title("Top 10 Opponents by Average Attendance")
plt.ylabel("Average Attendance")
plt.xlabel("Opponent")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



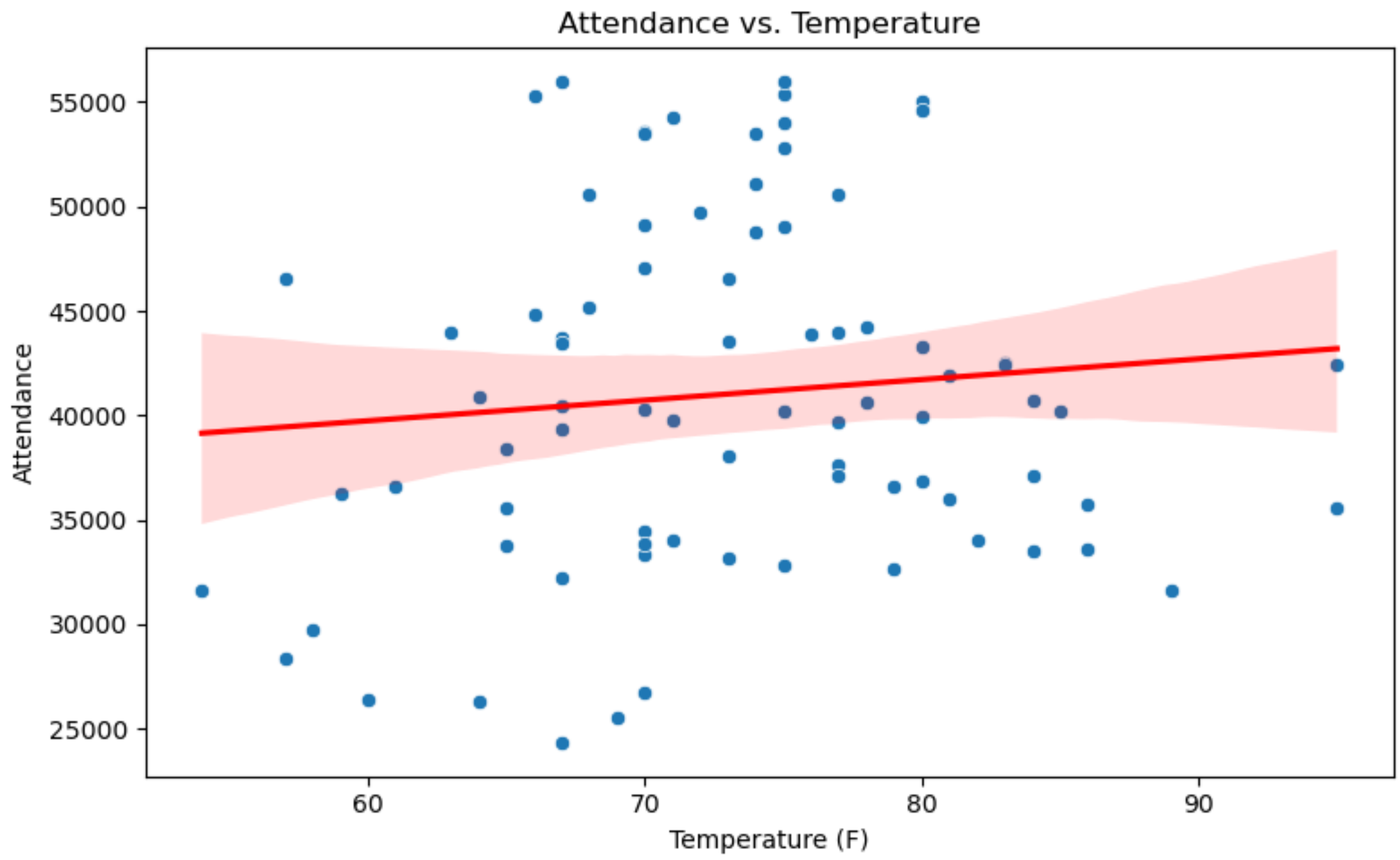



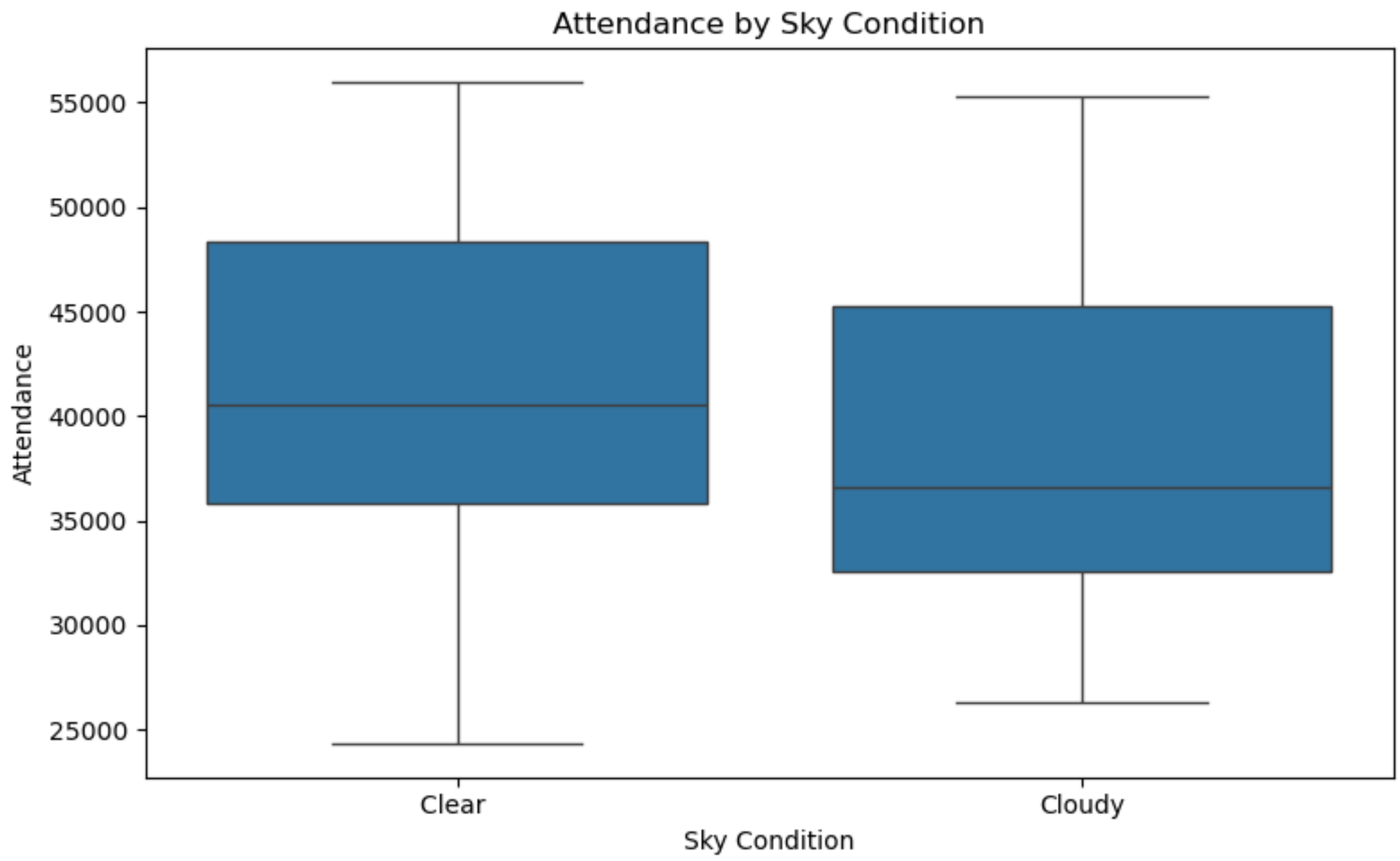

We see here that night games have slightly higher attendance on average, although not by much. The top 3 opponents by average attendance are the Angels, the Mets, and the Nationals.

## Weather Impact

In [37]:
```
plt.figure(figsize=(8, 5))
sns.scatterplot(x="temp", y="attend", data=dodgers_df)
sns.regplot(x="temp", y="attend", data=dodgers_df, scatter=False, color='red')
plt.title("Attendance vs. Temperature")
plt.xlabel("Temperature (F)")
plt.ylabel("Attendance")
plt.tight_layout()
plt.show()

plt.figure(figsize=(8, 5))
sns.boxplot(x="skies", y="attend", data=dodgers_df)
plt.title("Attendance by Sky Condition")
plt.xlabel("Sky Condition")
plt.ylabel("Attendance")
plt.tight_layout()
plt.show()
```




There is a slight positive correlation between temperature and attendance, suggesting that warmer days have higher attendance. More significant than this however is the difference in attendance between clear and cloudy days, with clear days having much higher average attendance.

## Linear Regression and Random Forest Regressor

In [ ]:
```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, r2_score
from sklearn.preprocessing import OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline

# Prepare features and target
X = dodgers_df.drop(columns=["attend", "day", "month"])
y = dodgers_df["attend"]

categorical_cols = X.select_dtypes(include=["object"]).columns.tolist()
numerical_cols = X.select_dtypes(include=["int64"]).columns.tolist()

preprocessor = ColumnTransformer([
    ("cat", OneHotEncoder(drop="first"), categorical_cols),
    ("num", "passthrough", numerical_cols)
])
```

```
lr_pipeline = Pipeline([
    ("preprocessor", preprocessor),
    ("model", LinearRegression())
])

rf_pipeline = Pipeline([
    ("preprocessor", preprocessor),
    ("model", RandomForestRegressor(random_state=42, n_estimators=100))
])

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

# Fit and predict
lr_pipeline.fit(X_train, y_train)
lr_preds = lr_pipeline.predict(X_test)

rf_pipeline.fit(X_train, y_train)
rf_preds = rf_pipeline.predict(X_test)

# Evaluation
lr_mae = mean_absolute_error(y_test, lr_preds)
lr_r2 = r2_score(y_test, lr_preds)

rf_mae = mean_absolute_error(y_test, rf_preds)
rf_r2 = r2_score(y_test, rf_preds)

print("Linear Regression - MAE:", round(lr_mae, 2), "R²:", round(lr_r2, 3))
print("Random Forest - MAE:", round(rf_mae, 2), "R²:", round(rf_r2, 3))
```

```
Linear Regression - MAE: 6828.43 R²: 0.278
Random Forest - MAE: 7493.96 R²: 0.217
```

Here I attempted to fit both a Linear Regression model and a Random Forest regressor to predict the attendance from the data. The Linear Regression model had an R2 of 0.278, and the Random Forest regressor had an R2 of 0.217, suggesting that neither model is a particularly good fit for predicting attendance.

## Recommendations

From our data analysis, we see that bobblehead giveaways have some of the greatest impact on game attendance. Because attendance tends to be lower on Mondays and Wednesdays, I would recommend doing bobblehead giveaways on those days to increase attendance on days that are usually slower. We also see in the data that shirt giveaways had a positive impact on attendance, so it may be a good idea to do shirt giveaways on Fridays as Fridays usually have the lowest attendance for the weekends. Another recommendation would be to do promotions for games against popular rivals in order to boost the already higher attendance for these games. Finally, I recommend doing more promotions or possibly discounted tickets for mid-week games that tend to have lower attendance.

## Source

Los Angeles Dodgers MLB Attendance Data - Provided on Blackboard by instructor https://cyberactive.bellevue.edu/ultra/courses/_538531_1/cl/outline